

Real-Time Anomaly Detection and Secure Intrusion Prevention: A Modular Approach

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Abstract—In the contemporary society the need to protect computer systems as well as the networks from anomalies, cyber threats and operational failures is inevitable. The project represents a novel multi-agent IDS intended to continually supervise system parameters, identify computer irregularities, and prevent potential attacks. The system is equipped with real time dynamics of monitoring, digital signature using the RSA technique and secure user authentication using JWT. IDS works on Multi-threaded model, when necessary parameters go out of limit, the IDS detects the deviation from normal behavior. This makes it alert on various incidents such as potential cyber-attacks or system overload, necessary cases which are protected by cryptographic hash. Its design involves testing to ensure it can handle various scenarios in real life, which helps avoid incidences wherein the system is down, and organizations within sectors such as healthcare and finance face lots of losses. It is designed to combine modern techniques with friendly interface providing efficient and secure approach to solve contemporary system monitoring problems at various scales.

Index Terms—Multi-Agent IDS, RSA Digital Signature, JWT Authentication, Cryptographic Hash

I. INTRODUCTION

Today, computer systems and networks must be secured as we run in this connected digital landscape. Technology has become an important aspect of organizations, businesses, as well as some individual users operations. However, as this is, they're vulnerable to a wide range of cyber threats, including unauthorized access, malware attacks, data breaches, and system anomalies. This has resulted in the development and deployment of automated solutions that automatically detect and respond to threats, so called Intrusion Detection Systems (IDS). The real world problem here is the detection of anomalies in system behavior well before they become big incidents. For instance the peaks in CPU usage, loading up on memory, or unauthorized access attempts, are typically preceding major systems failures or cyber attacks. Traditionally, the detection efficiency of such dynamic and evolving threats is not enough for real time scenarios. Furthermore, cyberattacks are growing more complex and require intelligent systems that can monitor and analyse and react against potential threats prior to occurring. The objectives of this project are to address these challenges by building a robust and intelligent IDS able to protect the computing environment. The implemented IDS is a multi agent system whose function is to monitor

system metrics such as CPU and memory, evaluates faults, and manages replication and recovery processes. The system is built to find anomalies in real time capturing predefined thresholds and notifying the administrators immediately for necessary action. The system employs cryptographic techniques, by which the alerts received are authentic and tamper-proof by leveraging RSA based digital signatures. In the real world where there could be malicious actors trying to take advantage of alerting mechanism itself, this feature is key. This project has the advantage of being able to fit real world use cases. Moreover, this IDS can detect system anomalies in expensive industries like finance, healthcare, and manufacturing, well before major downtimes and thereby incurring reasonable costs. In cybersecurity, detecting unusual behavior in system can be used as an early warning for cyber attack, so that quick mitigating measures can be taken if any incident occurs. As it integrates with JWT based authentication, critical metrics are only available to authorized users, and prevents unauthorized users from exploiting the monitoring system. The real time aspect of the project, along with its multi threaded architecture allow the system to run monitoring, fault evaluation, and replication processes at the same instance without degrading system performance. It also sends email alerts for critical anomalies respecting secure and user friendly SMTP with App Passwords. This project addresses real world problems of system failures, operational inefficiencies as well as cybersecurity threats which in turn provides a scalable, secure, and efficient solution to modern system monitoring needs. It shows how the forces of emerging technologies and methodologies are used to foster intelligent systems to address real world challenges that are reliable, secure, and continuous in this ever more digitized world. As an ongoing proactive defense mechanism, this system is designed not only to detect and react against threats but aid in the resilience of computing systems, to make it an invaluable tool for countering cyber risks and operational vulnerabilities.

II. LITERATURE SURVEY

Agoramoorthy[1] examines the evolution and growth of signature based intrusion detection systems (IDS) within hybrid framework. Kumar and Spafford early identified the efficiency of signature based systems for known threats but pointed to

the limitations of signature based systems for attacks which are polymorphic or novel. The only improvements to these systems which have been researched since are to augment them with anomaly detection, stateful protocol analysis, and machine learning to increase adaptability and performance. As solutions to dynamic threat environments, creating and updating automated signatures have been investigated. Previous research also focuses on database management and real time efficiency, including practical challenges for such a network and of high traffic networks, as well as advanced persistent threats. With continued increased interest in combining heuristic and behavioral with traditional signature based methods in an effort to boost the robustness and real time adaptation of IDS to counter effective sophisticated cyber threats, researchers have turned to this direction.

In addition, Sipahutar[2] describes the development and application of digital signature technology for boosting data security and integrity. Digital signatures, based on cryptographic algorithms, ensure data authenticity and integrity transmission through it is outlined. The foundational concepts underlying secure authentication through sequential cryptographic operations was first explored in early work, including that of Lamport's hash chain methodology. The subsequent work was extended to other techniques such as elliptic curve cryptography and Schnorr digital signatures to enhance computational efficiency and security. Digital signature chains – inspired by the blockchain like structure – have been understudied in fields like IoT and healthcare despite digital signatures are common in these fields. The chain based approach in place here implements chronological integrity and traceability to data transmissions in a way that resolves gaps of typical signature methods. Examples of challenges faced such as managing public keys are given, and opportunities for the future are explored including the use of shared public keys to improve the usability and scalability of this scheme.

In a work by Nanaware[3], we examine the evolution of email security measures that have been adopted to minimize attacks on phishing and spoofing sorts. Early protocols like STARTTLS, SPF, and DKIM all had their place in the security of email, but they also all had their limitations among them: lack of recipient action, spoofed IPs etc. Incorporated by Kucherawy and Zwicky DMARC is based on SPF and DKIM to enforce stronger domain authentication policies, and scale without third party externals. Other researchers focused on phishing awareness, or email classification systems to deal with spam and malicious emails. In fact, case studies of global email providers showed DMARC was effective against security and email deliverability threats. DMARC has been mandated by governments and institutions, such as the UK's National Cyber Security Centre and the US Department of Homeland Security, to beat phishing at scale. The second contribution from the paper is the automation of DMARC implementation and offering actionable insights with advanced analytics and real time reporting through DMARCBBox.

In this chapter, Cao[4] discusses the evolution and challenges of anomaly detection methods for industrial network

security. However, deterministic intrusion detection methods and feature based methods were able to scale and were accurate, but remained limited. The machine learning techniques, such as support vector machines (SVMs) and convolutional neural networks (CNNs), are introduced to improve anomaly detection accuracy and adaptability. Nevertheless, due to limitations in computational efficiency and time consumption in the high-dimensional and large scale industrial data, these methods can barely work. Practical limitations of the traditional and the deep learning models have been probed on recent research in which advanced dimensionality reduction techniques and semi-supervised learning have been central to improving both accuracy and real time applicability. Naturally, this is the context in which the proposed semi-supervisory anomaly detection approach is based on, where we use manifold learning to capture the data dimensionality before classification through the application of K-NN, aiming at achieving both computational efficiency and precision of the detection.

As a fundamental piece of network security, intrusion detection systems (IDS) are examined by Li[5]. Current systems such as firewalls and encryption rely on static passive approaches, and therefore are not effective at solving current security problems, with their internal threats and advanced attack patterns. IDS bridges the chasm between real time detection and response to internal and external threats. Research in the field of IDS has studied numerous methods, among them abuse detection, i.e., detecting malware by identifiers, and anomaly detection, i.e., detecting deviations from normal behavior, but such methods suffer from high rates of false positives and implementation complexities. Expert systems and honeypot technologies are integrated to enhance IDS capabilities, making their use and adaptive elements combined in a proactive and an adaptive defense. However, an expert system can reason in terms of the domain, but it is not robust to unknown attacks. The attackers are drawn by honeypots, gathered intelligence, updated knowledge base to enhance IDS' adaptability. These insights are built on by the proposed system which is a hybrid of expert systems and honeypots for an adaptive and self learning IDS.

As a major threat to cryptographic systems, RSA encryption and signatures, fault injection attacks are already evolving, and in this paper, Barbu[6] discusses it. The work begins with the foundational work by Boneh et al. in 1997, where they emphasize the need to examine cryptographic protocols for faults. Through the years, many fault models and techniques have been developed, including instruction skipping and random-byte faults, which show the possibility of breaking cryptosystems. In previous research, lattice reduction techniques, in particular LLL and BKZ, have been used to increase the effectiveness of fault attacks, with work on deterministic signature schemes. RSA-CRT implementations are used as the target of a seminal fault attack, the Bellcore attack, for key recovery. Based on these advances, the paper extends the practical reach of lattice based fault attacks against RSA encryption and signatures by proposing novel and relaxed fault

attacks on these schemes that allow much more error tolerance.

We review existing studies on token based authentication methods such as JSON Web Tokens (JWT) and Platform Agnostic Security Tokens (PASETO) as Nugraha[7] points out. Previous work has compared JWT performance under the algorithms of RSA, ECDSA, and HMAC with regards to token generation time, size, and transfer time. While companies conducted these studies, however, they often did not include robust security testing against common API vulnerabilities. PASETO has been investigated on its cryptographic robustness but most times, analysis on its performance or penetration testing has been scarce, and less is often been conducted to its performance or penetration testing, especially on OWASP API Security Top Ten 2019 vulnerabilities. This gap highlights the gap in understanding and demanding a holistic performance and security evaluation that encompasses both tokens in order to deliver actionable insights in secure RESTful API implementation.

Electronic signatures: existing methods and their related issues in the context of e-commerce are described by Zhu[8]. Due to the vulnerabilities in security and authenticity of the present manual signatures and seals, traditional manual signatures and seals remain inadequate for modern electronic commerce. Signatures and time stamping techniques are used for data integrity and authentication and digital watermarking for data concealment and anti tampering. Watermarking based on wavelet transforms and on the combination of public key infrastructure and digital signatures have been explored in previous studies. However, there exist many lack of complete integration of timestamping and watermarking for signature validation. Then, this paper develops a new electronic contract signature algorithm that combines digital signatures, time stamps and watermark in wavelet transform modulus maxima thus providing improved security, untraceable data integrity and tamper resistance for e-commerce contracts.

Rajeshwaran[9] established that cryptographic hash functions are an indispensable part of current information security as they help to check message authenticity and integrity. It covers the transformation from easy algorithms like MD5 and SHA1 to complex ones that are SHA2 and SHA3. The survey focuses CA's capabilities in the cryptographic application area because of its complexity and chaotic behavior. It was shown that prior works like that employing the hybrid CA rules and Omega-Flip networks, gained the deliberate avalanche effects. But these methods were not dynamic in their nature besides; they suffered from lack of efficient collision resistance. Based on these findings, this study proposes the CABHA, which incorporates non-linear CA rules and a specific transformation function to improve the randomization, diffusion, and collision resistance that are intrinsic to hash-function-based cryptographic techniques.

Park[10] investigates the use of machine learning and ambient intelligence in an intrusion detection system based in a smart factory. The system considerably improves anomaly detection rates by 33% and processing efficiency by 29% when compared to conventional methods by utilizing a hybrid

strategy of supervised and unsupervised learning to improve security threat identification. This system, which was installed at H Company, offered affordable diagnostics and decreased operating expenses by identifying problems such as the need for equipment replacement. With implications for wider deployment in smart factories and other industries, the suggested architecture shows how machine learning can be integrated for proactive threat management in complex industrial environments.

Almseidin[11] evaluates the performance of various machine learning methods for intrusion detection using the KDD dataset, which contains examples of various attack types such as DOS, R2L, U2R, and PROBE. Seven classifiers, including Decision Table, Random Forest, and Naive Bayes, were assessed. With the highest accuracy (93.77%) and lowest false positive rate, the Random Forest classifier was the most effective at reducing missed attacks, while the Decision Table classifier had the lowest false negative rate (0.002). However, in every statistic, no single classifier fared better than the others, highlighting the need for hybrid or tailored intrusion detection systems.

Musa[12] emphasizes how Intrusion Detection Systems (IDS) use anomaly-based and signature-based approaches to secure networks, with machine learning improving the detection of new threats. With more accuracy and fewer false positives, hybrid and ensemble classifiers perform better than single classifiers. Commonly used datasets are NSL-KDD, CIC-IDS2017, and UNSW NB-15; however, more recent datasets are advised. Through the removal of superfluous features, feature selection enhances IDS performance. Enhancing zero-day attack detection, scalability for big datasets, and making sure models generalize well across various datasets are some of the issues of the future.

ELHejazi[13] suggests a safe and effective way to use JSON Web Tokens (JWT), OpenID, and OAuth 2.0 to improve the security of Software-Defined Networking (SDN) controller REST APIs. It draws attention to flaws in conventional SDN controllers that are vulnerable to injection attacks and illegal access because they lack strong authentication. JWT-based token authentication is used in the suggested approach to provide fine-grained access control and safe user verification. Tests conducted in a Cisco DNA sandbox environment show that the method improves API session reliability, decreases human labor, and automates API authentication. The approach offers comprehensive session data and lowers the danger of token misuse, indicating a notable increase in SDN API security and speed. The goal of future research is to combine the solution with detection techniques to thwart DDoS attacks.

Gauri[14] gives a review which places special emphasis on developments in hardware accelerators for BLAKE2b and other cryptographic hash algorithms. Although it is computationally intensive for big inputs, key papers emphasize its effectiveness in secure applications like blockchain and digital signatures. Current studies investigate FPGA-based acceleration, which makes use of pipelining and parallel processing, to lower latency and increase throughput. The suggested architec-

ture, a fully pipelined BLAKE2b hardware accelerator built on FPGA, overcomes conventional software-based constraints and exhibits notable performance improvements with lower latency and increased computational efficiency. Future developments will concentrate on area, power, and integration into larger cryptographic systems.

Chen[14] suggests a data mining-based network intrusion detection solution for power monitoring systems in order to overcome the lengthy detection timeframes and poor accuracy of conventional approaches. Utilizing data mining methods and clustering partitioning to preprocess intrusion data, the technique improves the identification of anomalous data in network traffic. For 500 data groups, simulation tests showed that the suggested method outperforms conventional techniques by 15.6% in terms of detection accuracy, 12.3% in terms of false alarm rates, and runs about 1.8 seconds faster. The safe and steady operation of electricity systems can be guaranteed with this effective and dependable technology.

III. METHODOLOGY

When it comes to IDS implementation, real time monitoring, deviation detection as well as sending alerts of a system are suggested to try and protect a system against the possible threats. The system is designed in modular fashion and has agents for monitoring of the system, assessment of faults and replication/recovery solutions. Built on a modular architecture, the system includes agents to monitor system performance, evaluate faults, and manage replication and recovery. To improve security it includes RSA based digital signatures, JWT Authentication to endpoints API and alerting with mails.

A. System Architecture

IDS is a modular architecture with each component being tasked for some responsibilities. This design provides scalability, flexibility, and competent resource management.

- *SystemSentinelAgent*: Anomaly detection is performed by the SystemSentinelAgent which monitors system performance metrics.
Data Collection: Key metrics such as the CPU utilisation and memory usage are gathered using the psutil library. Threshold Analysis: If predefined thresholds are exceeded (e.g. CPU usage >10%, memory usage = 75%), their anomalies are flagged. Alert Mechanism: The agent generates a detailed alert upon detecting an anomaly, which he digitally signs using RSA encryption and send via email to the administrator.
- *SystemFaultEvaluationAgent*: This agent evaluates the data, collected by the SystemSentinelAgent and identifies potential faults or vulnerabilities. It tracks patterns and trends and flags recurring issues, logging findings for review to give administrators actionable insight into what's happening in their digital world.
- *SystemReplicationAgent*: This agent monitors for degradation or faults to help ensure continuity with system. It ensures system states or replicates critical components

in order to maintain system stability when anomalies or failures occur.

- *ProfileDatabase*: The normal system behavior is maintained in the form of updated profiles and this Profile-Database is the base for the anomaly detection. It is a dynamic acquisition which integrates new data and helps to locate deviations and raise the accuracy of detection.
- *Local System Intelligence Agent (LSIA)*: All agents are coordinated by the LSIA and their information is utilized so as to eliminate redundancy and maximize system performance. This helps to smooth out the workflow by ensuring that components can speak easily to one another.

B. Alert and Anomaly Detection

The IDS depends on anomaly detection mechanism, it provides real time monitoring and response.

- *Detection Algorithm*: The algorithm looks at observed metrics and compares them against predefined thresholds in case of anomalies.
- *Alert Generation*: Alerts come with timestamps, full text descriptions, and pertinent metrics. It is digitally signed for authenticity.
- *Email Notification*: SMTP with app password authentication is used to send secure email notifications.

C. Data Security

The IDS is robust insecurity features for safeguard system integrity and secure sensitive data.

- *JWT Authentication*: JSON Web Tokens (JWT) are used to secure API endpoints by checking to make sure that the token that is included is valid. Tokens are issued, post authentication and expire a set amount of time later (30 minutes, for instance).
- *Digital Signatures*: The alert messages are signed with RSA encryption to prove both their authenticity and integrity. These can later be verified from public keys.

D. Real-Time Dashboard and APIs

Monitoring and control of the IDS are offered through a user friendly web interface and API endpoints.

- *Dashboard*: The dashboard is built with Flask, displaying real time metrics including CPU, Memory and a recent anomaly log.
- *API Endpoints*:
/login: Once users have authenticated themselves, it issues JWT tokens.
/metrics: It offers current metrics (system metrics, anomaly details) complete only for a valid token.

E. Web-Based Monitoring and Response Workflow

Unlike a command and control IDS, the IDS runs in continuous loop to efficiently detect, notify, analyze, and recover.

- *Monitoring*: The agents continue to collect metrics continuously.
- *Detection*: Violations of the threshold are flagged as anomalies.

- *Notification*: They are notified of events through alerts, and signed appropriately.
- *Analysis*: The `SystemFaultEvaluationAgent` performs fault pattern and vulnerability analysis.
- *Recovery*: When necessary, the `SystemReplicationAgent` will start performing recovery actions in order to keep the system stable.

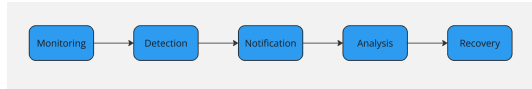


Fig. 1. System Flow

F. Scalability and Extensibility

Modularity of the IDS allows its design to be adapted easily to changing requirements.

- *Adding New Agents:* Specific tasks can be handled by additional agents, as well as new metrics of your application.
- *Customizing Thresholds:* Specific environments provide direction on thresholds and detection algorithms.
- *Extending Functionality:* Integration with external systems in a broader operational scope is possible with secure APIs.

G. Implementation Technologies

The IDS is implemented in Python, with its core functionality leveraged by several libraries.

- *psutil*: To collect collection system metrics.
- *smtplib*: As it is related for sending email alerts securely.
- *jwt*: To handle authentication and managing tokens.
- *cryptography*: It is used in RSA encryption, or better yet, for digital signatures.
- *Flask*: Now is used for building the dashboard, and APIs.

IV. RESULTS

- **Real-time Monitoring and Detection:** Real time detection of anomalies on real system metrics. Potential security threats, alerts timely.

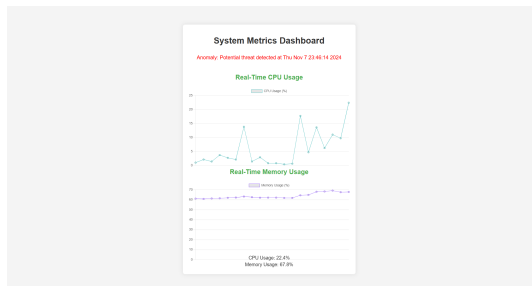


Fig. 2. Real-time Monitoring

- **Digital Signature:** Digitally signed alerts prevents access of unauthorized parts to system measurements and control.

[illegible]

Fig. 3. Digital Signature

- **Automated Email Response:** Email notification of anomalies ‘immediately’. This reduces the manual intervention with automated fault analysis.

[illegible]

Fig. 4. Email Response

- **Token or Real Time Metrics:** The CPU and Memory usage metrics are taken from the backend and sent to the frontend for graph plotting and real time visualialization

```
oken is valid
{'cpu': 0.8, 'memory': 61.4, 'anomaly': None}
127.0.0.1 - - [07/Nov/2024 23:44:31] "GET /metrics HTTP/1.1" 200 -
Analyzing system data...
Managing replication and recovery...
Analyzing system data...
Analyzing system data...
```

Fig. 5. Realt time Metrics being sent

- **Authentication:** JWT based authentication for secure communication. Before entering into the application, the user have to authenticate themselves so as to prove his/her identity. Hence a login page is shown at first after opening the web application.

Login

Login

Fig. 6. Login Page

CONCLUSION

The proposed Intrusion Detection System (IDS) can be composed as a robust framework for real time anomaly detection, secure communication and system stability. The system combines SystemSentinelAgent for anomaly detection, SystemFaultEvaluationAgent for fault analysis, and SystemReplicationAgent for recovery to continuously monitor, and

deliver prompt responses to anomalies. Further security added in includes RSA digital signatures and JWT authentication to protect sensitive data and build trust in alert mechanisms. The real time dashboard and secure API endpoints also give the administrators a simple yet powerful interface to monitor system performance and take action whenever incidents occur. Its flexibility and extensibility allow this system to fit any environment or changes required. Under such circumstances, its flexibility to accept more agents, to set thresholds, and to integrate with external systems makes it a desirable single solution for most modern intrusion detection problems. It adds security measures such as JWT for authentication, digital signatures for determining the integrity of the alerts, secure email transmission via OAuth 2.0 and an IDS for detecting potential threats. For a web security in e-commerce platforms which includes data integrity, user authentication and system availability, these are highly relevant.

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